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The Effects of Valsartan and Captopril on Reducing Microalbuminuria in Patients with Type 2 Diabetes Mellitus: A Placebo-Controlled Trial

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ABSTRACT

This multicenter, randomized, double-blind, placebo- and captopril-controlled, parallel-group trial evaluated the efficacy and safety of valsartan 80 and 160 mg in patients with incipient diabetic nephropathy. Patients were randomized to receive either valsartan 80 mg or 160 mg once daily, captopril 25 mg 3 times daily, or placebo. The study comprised 122 normotensive and treated hypertensive patients with type 2 diabetes mellitus and microalbuminuria (mean age, 56 years; 90% white). Treatment lasted 52 weeks. Efficacy variables included albumin excretion rate (AER), progression to clinical proteinuria, and glomerular filtration rate. In both the valsartan 80-mg (n = 31) and 160-mg (n = 31) groups and in the captopril group (n = 29), a decrease in AER from baseline was observed at end point, compared with an increase in the placebo group (n = 31). The positive effect of valsartan 80 mg versus placebo on AER was statistically significant (95% confidence interval [CI] for end point/baseline ratio: 0.365 to 0.966); the 95% CI for valsartan 160 mg versus placebo was 0.407 to 1.043. No significant differences in AER occurred in the comparisons of valsartan 80 mg and valsartan 160 mg versus captopril. The percentage of patients with trial drug-related adverse experiences was highest in the captopril group (34.5%). The corresponding values for the groups receiving valsartan 80 mg, valsartan 160 mg, and placebo were 9.7%, 22.6%, and 13.8%, respectively. Results of this study suggest that treatment with valsartan slows the progressive rise of AER in normotensive and treated hypertensive patients with type 2 diabetes mellitus with comparable efficacy and superior tolerability to captopril. *Key words:* valsartan, type 2 diabetes mellitus, microalbuminuria, albumin excretion rate.

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INTRODUCTION

The renin-angiotensin system (RAS) plays a role in the pathophysiology of diabetic nephropathy, a clinical syndrome characterized by persistent albuminuria, a progressive decline in glomerular filtration rate (GFR), and hypertension. Several studies in patients with type 1 or 2 diabetes mellitus have demonstrated that treatment with angiotensin-converting enzyme (ACE) inhibitors significantly slows the progression of diabetic nephropathy to more advanced stages, as well as delays the onset of overt proteinuria.[1–5] Although ACE inhibitor therapy is effective in the treatment of diabetic patients with early nephropathy even in the absence of hypertension, these agents inhibit the RAS relatively nonspecifically, resulting in the characteristic side effects of persistent dry cough and—rarely—life-threatening angioneurotic edema.[6]

Valsartan is a new, specific, potent, orally active angiotensin II receptor blocker. Angiotensin receptor blockers competitively block the AT_1 receptor subtype, which mediates all the known cardiovascular effects of angiotensin II. This receptor is particularly prominent in the renal tubules and vasculature and modulates renal blood flow, GFR, tubular epithelial transport, renin release, and cell growth.[7] Because animal models of renal disease have shown that the effects of ACE inhibitors and angiotensin receptor blockers are comparable,[8,9] angiotensin receptor blockers would be expected to be as effective as ACE inhibitors in the treatment of diabetic nephropathy in humans. A recent study[10] in patients with nondiabetic renal disease confirmed that losartan and the ACE inhibitor enalapril were equally effective in decreasing urinary protein excretion.

The aim of the present study was to compare the efficacy and safety of 52 weeks of treatment with valsartan (80 and 160 mg once daily) with that of placebo and captopril 25 mg 3 times daily on progression of urinary albumin excretion and renal function in normotensive and treated hypertensive patients with type 2 diabetes mellitus and nephropathy.

PATIENTS AND METHODS

Male and female outpatients ≥18 years of age, of any racial background with type 2 diabetes mellitus and incipient diabetic nephropathy were eligible for random allocation into the trial. Incipient diabetic nephropathy was defined as an albumin excretion rate (AER) between 20 and 300 μg/min with GFR ≥60 mL/min per 1.73 m^2 at visit 1.

Normotensive and treated hypertensive patients with a sitting diastolic blood pressure (SDBP) of ≤95 mm Hg and a sitting systolic blood

pressure (SSBP) ≤160 mm Hg were included in the study. Women of childbearing potential were included only if they were using an effective method of birth control not based on combined estrogen/progestogen. Patients being treated with ACE inhibitors or calcium channel blockers were included provided they discontinued treatment for at least 28 days before randomization (visit 2).

Patients with "brittle" diabetes—that is, at increased risk of hypoglycemia—or a history of noncompliance with medical regimens were excluded from the study. Patients who experienced symptomatic hypotension (SDBP <60 mm Hg), who progressed to hypertension (SDBP >95 mm Hg despite treatment with beta-blockers or diuretics), or who experienced serious adverse experiences (AEs) related to the study treatment were discontinued from the trial.

The study was approved by the appropriate institutional review board and was conducted according to good clinical practice requirements. All patients provided written informed consent before entering the study.

Study Design

This multicenter, randomized, double-blind, placebo- and active-controlled, parallel-group trial was conducted in 4 centers in Canada. After a 28-day washout period for patients receiving ACE inhibitors or calcium channel blockers, patients were randomly allocated to 1 of the 4 treatment groups in a 1:1:1:1 ratio. They received either valsartan 80 mg once daily, valsartan 160 mg once daily, captopril 25 mg 3 times daily, or placebo. Captopril was selected as an active control after its approval for use in diabetic nephropathy in patients with type 1 diabetes mellitus and was given at the dose proven to attenuate progression of proteinuria.[1] To maintain blinding, all trial medication was identical in size, color, appearance, and taste. Treatment lasted 52 weeks for all groups.

Glycemic control was maintained during the study by means of the patients' customary treatment. Use of antihypertensive medication (except diuretics or beta-blockers), estrogen replacement therapy, or thyroid medication <6 months before entry into the trial was prohibited.

Patients were assessed clinically at baseline and at 6, 12, 26, 38, and 52 weeks of treatment. AER and GFR were assessed at baseline and after 12, 26, 38, and 52 weeks of treatment. AER was measured from a 24-hour urine sample by means of a radioimmunometric assay (Pharmacia Albumin RIA, Kabi, Pharmacia, Sweden), and GFR was determined by measuring the clearance of ^{99}Tc DTPA. Systolic and diastolic blood pressures were measured at each visit in sitting and standing positions, according to World Health Organization guidelines.

The primary efficacy variable was the change in AER at end point (after 52 weeks of treatment or at premature discontinuation). The num-

ber of patients progressing to clinical proteinuria (2 AER values >300 μg/min within 6 months after baseline) was also determined. The secondary efficacy variable was GFR after 52 weeks of treatment or at discontinuation.

Safety and tolerability were primarily evaluated from the frequency of adverse experiences. Other safety variables included changes in routine laboratory test results and blood pressure readings.

Statistical Analysis

The sample size was calculated based on change in AER over the 52-week study. Using Dunnett's adjustment for the 2 tests of primary interest (valsartan 80 mg or 160 mg vs placebo), 28 assessable patients per group were calculated to be required to provide 80% power at the 0.05 significance level. A sample size of 120 randomized patients (30 patients per group) allowed for possible dropouts.

The efficacy evaluations were based on the analysis of the intent-to-treat population. This included all randomized patients with at least 1 postbaseline measurement for the respective efficacy variable.

The primary efficacy variable, change in AER from baseline to end point or discontinuation (last observation carried forward), was expressed as the ratio of mean AER at end point to mean AER at baseline (end point/baseline ratio). This was analyzed using a general linear model fitting baseline, treatment, and center effect in the main analysis. Because of skewed distribution, all AER values were transformed to their logarithms before being analyzed in the model. Dunnett's procedure for multiple comparisons was used to show statistical significance compared with placebo.

The safety analysis was descriptive and included all randomized patients who had received at least 1 dose of study medication.

RESULTS

One hundred twenty-two patients were randomly allocated into the valsartan 80-mg group (n = 31), the valsartan 160-mg group (n = 31), the captopril group (n = 29), or the placebo group (n = 31). Of the 122 patients randomized, 103 (84.4%) completed the study. Of the 19 patients (15.6%) who discontinued treatment prematurely, 7 (22.6%) were in the valsartan 80-mg group, 1 (3.2%) was in the 160-mg group, 4 (13.8%) were in the captopril group, and 7 (22.6%) were in the placebo group. Of the patients who discontinued treatment prematurely, 4 patients (1 in the valsartan 80-mg group, 1 in the valsartan 160-mg group, and 2 in the captopril group) did so because of AEs, and 2 (1 in the valsartan 80-mg group and 1 in the placebo group) because of abnormal laboratory test results. The remainder discontinued because of failure to meet the protocol criteria

(1 patient), noncompliance (7 patients), withdrawal of consent (4 patients), or administrative problems (1 patient).

Table I summarizes the baseline demographics by treatment group. All groups had a preponderance of men. The proportion of women ranged

Table I. Summary of demographic and baseline data (all randomized patients).

	Valsartan 80 mg (n = 31)	Valsartan 160 mg (n = 31)	Captopril (n = 29)	Placebo (n = 31)
Sex, no. (%)				
Female	9 (29.0)	13 (41.9)	8 (27.6)	3 (9.7)
Male	22 (71.0)	18 (58.1)	21 (72.4)	28 (90.3)
Race, no. (%)				
White	27 (87.0)	31 (100.0)	24 (82.8)	28 (90.3)
Black	0 (0)	0 (0)	1 (3.4)	0 (0)
Asian	2 (6.5)	0 (0)	2 (6.9)	1 (3.2)
Other	2 (6.5)	0 (0)	2 (6.9)	2 (6.5)
Age (y)*	53.7 ± 9.5	58.3 ± 9.5	56.7 ± 10.0	55.5 ± 11.3
Body weight (kg)*	97.8 ± 20.2	96.7 ± 25.0	89.1 ± 16.7	93.6 ± 18.7
Antihypertensive medication use, no. (%)				
Yes	10 (32.3)	9 (29.0)	11 (37.9)	17 (54.8)
No	21 (67.7)	22 (71.0)	18 (62.1)	14 (45.2)
Blood pressure (mm Hg)*				
SSBP	135.0 ± 12.4	135.2 ± 12.8	136.8 ± 15.4	135.4 ± 17.5
SDBP	84.2 ± 7.5	80.7 ± 7.7	83.9 ± 6.0	83.5 ± 8.3
AER (μg/min)†	60.5	58.1	40.9	64.0
GFR (mL/min per 1.73 m^2)†	101.5	83.1	88.1	86.7

SSBP = sitting systolic blood pressure; SDBP = sitting diastolic blood pressure; AER = albumin excretion rate; GFR = glomerular filtration rate.
* Mean ± SD.
† Geometric mean.

from 9.7% in the placebo group (3/31) to 41.9% in the valsartan 160-mg group (13/31). The differences in sex ratio are unlikely to be relevant to the efficacy and safety results and conclusions of the trial. Treatment groups were essentially comparable for ethnic group (90% white overall), age, body weight, antihypertensive medication, SSBP, SDBP, and GFR. Baseline AER was somewhat lower in the captopril group than in the other 3 treatment groups (geometric mean in the captopril group, 40.9 μg/min vs 60.5 μg/min in the valsartan 80-mg group, 58.1 μg/min in the valsartan 160-mg group, and 64.0 μg/min in the placebo group).

Efficacy

In the valsartan 80-mg and 160-mg groups, as well as in the captopril group, AER decreased from baseline to end point, compared with an increase in the placebo group (Table II). Table III shows the comparisons between treatments in the end point/baseline ratio in AERs. All active

Table II. Geometric means of change from baseline in albumin excretion rates (AERs) (μg/min) for the intent-to-treat population.*

	Valsartan 80 mg (n = 27)	Valsartan 160 mg (n = 31)	Captopril (n = 29)	Placebo (n = 28)
Baseline	60.0	58.1	40.9	63.3
End point	43.3	45.8	30.1	74.8
End point/baseline ratio	0.72	0.79	0.73	1.18

* Intent-to-treat population included all randomized patients with at least 1 postbaseline AER measurement.

treatment groups significantly reduced AER compared with placebo ($P < 0.05$). After applying Dunnett's adjustment for multiple comparisons, the effect of valsartan 80 mg versus placebo remained statistically significant; for valsartan 160 mg versus placebo, however, the upper 95% adjusted confidence limit was above 1 (95% CI, 0.407 to 1.043). Compared with captopril, there were no statistically significant differences in reduction in AER for valsartan 80 mg (95% CI, 0.681 to 1.612) or valsartan 160 mg (95% CI, 0.760 to 1.743).

One patient (3.7%) in the valsartan 80-mg group, none in the valsartan 160-mg group, 1 (3.4%) in the captopril group, and 3 (10.7%) in the placebo group progressed to clinical proteinuria.

No significant increase or decrease from baseline in GFR occurred in any of the treatment groups (Table IV). There was also no significant effect

Table III. Contrast* between treatments for the end point/baseline ratio in albumin excretion rates for the intent-to-treat population.

Contrast	Contrast Mean	95% CI	95% CI Adjusted for Multiple Comparison
Tests of primary interest			
Valsartan 80 mg versus placebo	0.593	0.386 to 0.911 $P = 0.018$	0.365 to 0.966
Valsartan 160 mg versus placebo	0.652	0.431 to 0.986 $P = 0.043$	0.407 to 1.043
Tests of secondary interest			
Captopril versus placebo	0.566	0.370 to 0.868 $P = 0.009$	
Valsartan 80 mg versus captopril	1.048	0.681 to 1.612 $P = 0.831$	
Valsartan 160 mg versus captopril	1.151	0.760 to 1.743 $P = 0.503$	

CI = confidence interval.
* The contrasts estimate the ratio of the treatment means for the end point/baseline ratio of the respective variable. A value ≤1 means that the first treatment in the contrast is more likely to lower the value of the variable.

Table IV. Geometric means of change from baseline in glomerular filtration rates (GFRs) (mL/min per 1.73 m^2) for the intent-to-treat population.*

	Valsartan 80 mg (n = 28)	Valsartan 160 mg (n = 31)	Captopril (n = 28)	Placebo (n = 27)
Baseline	102.4	83.1	89.5	83.2
End point	95.0	74.3	89.9	76.8
End point/baseline ratio	0.927	0.894	1.005	0.923

* Intent-to-treat population included all randomized patients with at least 1 postbaseline GFR measurement.

of the 2 valsartan treatments on GFR compared with placebo ($P = 0.964$ and $P = 0.703$ for valsartan 80 mg and 160 mg vs placebo, respectively). None of the comparisons of secondary interest showed statistical significance.

Tolerability and Safety Profile

One hundred twenty patients were included in the safety analysis. Two patients from the placebo group who dropped out immediately after randomization were excluded.

The total number of patients with AEs (irrespective of relationship to study medication) was highest in the captopril group and lowest with valsartan 80 mg. Twenty-five patients receiving valsartan 80 mg (80.6%),

Table V. Summary of the number (%) of patients with trial drug–related adverse experiences (AEs).

	Valsartan 80 mg (n = 31)	Valsartan 160 mg (n = 31)	Captopril (n = 29)	Placebo* (n = 29)
Total patients with ≥1 trial drug–related AE	3 (9.7)	7 (22.6)	10 (34.5)	4 (13.8)
Dry cough	1 (3.2)	3 (9.7)	6 (20.7)	1 (3.4)
Diarrhea	0 (0)	1 (3.2)	1 (3.4)	0 (0)
Dizziness	0 (0)	0 (0)	3 (10.3)	1 (3.4)
Dyspepsia	0 (0)	0 (0)	1 (3.4)	0 (0)
Gastrointestinal disorder	1 (3.2)	0 (0)	0 (0)	0 (0)
Headache	0 (0)	1 (3.2)	0 (0)	1 (3.4)
Postural hypotension	1 (3.2)	0 (0)	0 (0)	0 (0)
Migraine	0 (0)	0 (0)	0 (0)	1 (3.4)
Nausea	0 (0)	1 (3.2)	0 (0)	1 (3.4)
Pyuria	1 (3.2)	0 (0)	0 (0)	0 (0)
Upper respiratory tract infection	0 (0)	0 (0)	1 (3.4)	0 (0)
Vertigo	0 (0)	0 (0)	1 (3.4)	0 (0)
Abnormal vision	0 (0)	1 (3.2)	0 (0)	0 (0)

* Two patients from the placebo group who dropped out immediately after randomization were excluded from analysis.

27 patients receiving valsartan 160 mg (87.1%), 28 patients receiving captopril (96.6%), and 24 patients receiving placebo (82.8%) experienced AEs. The high frequency of AEs even in the placebo group reflects the underlying health status of the patient group participating in this study.

Table V summarizes the AEs considered by the investigators to be related to trial drug. The highest percentage occurred in the captopril group (34.5%). The corresponding values for the valsartan 80-mg group, the valsartan 160-mg group, and the placebo group were 9.7%, 22.6%, and 13.8%, respectively. The excess in the captopril group was accounted for by a markedly higher incidence of dry cough and dizziness compared with valsartan or placebo (cough: captopril, 20.7%; valsartan 80 mg, 3.2%; valsartan 160 mg, 9.7%; placebo, 3.4%; dizziness: captopril, 10.3%; valsartan groups, 0%; placebo, 3.4%). No drug-related serious adverse events or deaths occurred during the study.

Four patients discontinued therapy prematurely because of AEs (1 aneurysm/cerebrovascular disorder in the valsartan 80-mg group, 1 uncontrolled hypertension in the valsartan 160-mg group, 1 blot hemorrhages and 1 dry cough in the captopril group). Two patients discontinued therapy because of abnormal laboratory values (1 patient receiving valsartan 80 mg had increased urea and decreased GFR and creatinine clearance, and 1 patient receiving placebo had increased AER, anemia, and elevated potassium values). No clinically relevant pattern in routine laboratory test results was observed in the 4 treatment groups for hematocrit, red blood cell count, platelet count, white blood cell count, creatinine, urea (blood urea nitrogen), chloride, and sodium.

Slight changes in SSBP and SDBP from baseline in the valsartan and captopril groups were observed but were not tested for statistical significance. The change from baseline to end point in SSBP was –2.4 mm Hg with valsartan 80 mg, +4.9 mm Hg with valsartan 160 mg, +1.2 mm Hg with captopril, and +2.9 mm Hg with placebo. The change from baseline to end point in SDBP was –5.2 mm Hg with valsartan 80 mg, –3.0 mm Hg with valsartan 160 mg, –3.2 mm Hg with captopril, and –1.4 mm Hg with placebo. The mean increase in SSBP at end point in the valsartan 160-mg group was mainly due to an extremely high blood pressure value in a single patient who was then withdrawn because of uncontrolled hypertension.

No trend in differences in glycemic control was observed between groups ([mean ± SD glucose values] valsartan 80 mg: baseline = 11.0 ± 3.48 mmol/L, end point = 10.5 ± 3.19 mmol/L; valsartan 160 mg: baseline = 10.6 ± 4.01 mmol/L, end point = 10.2 ± 3.96 mmol/L; captopril: baseline = 10.1 ± 3.37 mmol/L, end point, 9.92 ± 3.66 mmol/L; placebo: baseline = 10.5 ± 4.30 mmol/L, end point = 11.1 ± 3.81 mmol/L). Small changes from baseline were observed in glycosylated hemoglobin A_{1c} in all groups (valsartan 80 mg, –0.02% ± 1.61%; valsartan 160 mg, –0.63% ± 1.68%; captopril, 0.16% ± 2.05%; and placebo, 0.50% ± 2.02%).

DISCUSSION

Microalbuminuria is often a predictor of diabetic nephropathy in type 1 diabetes mellitus.[11–13] In normotensive patients with type 1 or type 2 diabetes and microalbuminuria, early treatment with ACE inhibitors has been shown to be effective in reducing AER.[1–5,14] This effect is thought to be independent of the reduction in systemic blood pressure.[4,15] Decrease in the permselectivity of the glomerular basement membrane and selective reduction of intraglomerular pressure have been suggested as the mechanisms of ACE inhibitor nephroprotection.[16–18]

The present study examined the effect of treatment with valsartan 80 mg, valsartan 160 mg, and captopril on AER and GFR in normotensive and treated hypertensive patients with type 2 diabetes mellitus and incipient diabetic nephropathy. Valsartan appears to have an efficacy profile similar to that of captopril for reduction in AER, with a better tolerability profile (ie, lower incidence of dry cough). Valsartan 80 mg, valsartan 160 mg, and captopril all resulted in a decrease in AER over time, whereas a slight increase was observed in the placebo group. After Dunnett's adjustment for an overall 5% level, the relative change in AER from baseline with valsartan 80 mg versus placebo showed a significant positive effect of valsartan treatment (95% CI, 0.365 to 0.966). No significant differences were observed in the comparisons of either valsartan group against captopril. Captopril showed a significant positive effect in comparison with placebo ($P = 0.009$).

Although the baseline AER was slightly lower in the captopril group than in the other 3 treatment groups, the analyses were not affected because the baseline value was included as an effect in the primary analyses, and the tests of primary interest compared the 2 valsartan groups with placebo.

Fewer patients in the active treatment groups (1 patient with valsartan 80 mg, none with valsartan 160 mg, 1 with captopril) progressed to clinical proteinuria compared with the placebo group (3 patients). These findings support the positive renal effects as measured by AER. However, because the total number of patients progressing to proteinuria was small ($n = 5$), no firm conclusions can be drawn.

All trial medications were generally well tolerated. The overall incidence of AEs assessed by the investigators as at least possibly related to study medication was highest in the captopril group (34.5%), compared with 9.7% in the valsartan 80-mg group, 22.6% in the valsartan 160-mg group, and 13.8% in the placebo group. The incidence of trial drug–related dry cough was also highest in the captopril group.

One possible explanation of the effect of active treatment on AER is simply lowering of blood pressure. Thus the primary linear model on AER was repeated to include the additional effects of change from baseline in

mean SDBP and mean SSBP at baseline. After Dunnett's adjustment, the effect of change from baseline in either SSBP or SDBP on AER outcome had no significant effect for valsartan 80 mg or 160 mg compared with placebo. In addition, because the treatment groups were comparable for concomitant hypertensive medications during the trial, it is reasonable to conclude from this study that angiotensin II receptor blockers, such as valsartan, exert similar beneficial effects on AER as ACE inhibitors, independent of blood pressure lowering.

Although the use of a balanced placebo group in a 1-year study in this patient population could raise an ethical question today, at the time this study was conducted (first patient enrolled in 1995), ACE inhibitor therapy was not yet established as the preferred therapy for patients with type 2 diabetes mellitus and microalbuminuria. The results of this study further strengthen recent evidence that type 2 diabetic patients with microalbuminuria require treatment with agents that can slow the progression of nephropathy.

CONCLUSION

Results of the present study suggest that valsartan slows the progressive rise of AER in normotensive and treated hypertensive patients with type 2 diabetes mellitus as effectively as captopril, but with better tolerability and a lower incidence of dry cough.

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